



Ideal L2 Self, Visual Learning Styles, and L2 Self Confidence in predicting Language Proficiency and L2WTC: A Structural Equation Modeling

以理想我、視覺學習型態、自信心來預測第二語言能力和溝通意願:結構方程模式之研究

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Published online: 26 September 2018

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Abstract

This study intends to identify the interrelationships between motivation, perceptual learning styles, cognitive variables, and language proficiency in a single framework. In particular, the role of the Ideal L2 self, visual learning styles, and L2 confidence for predicting learners' language proficiency and their willingness to communicate in real life were investigated. The results revealed that visual learning styles have a positive effect on the Ideal L2 self and are indirectly associated with L2 proficiency through the mediating function of motivation. These relationships provide evidence that learners with more vivid images of the distance between their actual and Ideal L2 selves find learning a goal-oriented activity and as a result, are encouraged to make more endeavors to reach higher levels of proficiency. The desire to bridge the gap between their current and ideal selves helps learners reduce anxious arousals and feel more competent to accomplish L2 tasks, which in turn helps learners be more prepared and willing to communicate. Finally, implications for language teachers and policy makers to inspire learners to become more engaged in L2 activities are discussed.

摘要

本研究旨在確定於單一架構下,動機、察覺學習類型、認知變數和語言能力之間的相互關係。特別調查以理想我、視覺學習型態、自信心來預測學習者的語言能力以及他們實際的溝通意願。研究結果顯示透過動機的中介效果,視覺學習型態對理想我有正向的影響以及與第二語言能力有間接的關聯。這些關係證明學習者有實際我和理想我之間距離的生動影像;他們發現學習是一個目標為導向的活動,因此受到鼓勵,進而更努力達到更高水平的語言能力。渴望彌合在實際我與理想我之間的差距,可以幫助學習者降低焦慮的產生,有助於學習者感到更有能力去完成語言任務;進而幫助他們做更充分的準備和有意願去進行溝通。本文針對語言教師和政策制定者,提供相關激勵學習者在投入語言活動的研究建議。

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Keywords Ideal L2 self · WTC · Visual learning styles · L2 communicative competence · Perceptual learning styles

關鍵詞 想我 · 溝通意願 · 視覺學習型態 · 第二語言溝通能力 · 察覺學習類型

Introduction

Individual differences, or having a personal trait feature, are attributed to the differences among second/foreign language learners in terms of their demographic background including ethnicity, gender, cognitive styles, nationality, learner's beliefs, learning styles, and age. Of all the individual differences associated with L2 learning, motivation is deemed as one of the most crucial factors with a significant function in achievement (Lanvers 2016). It is not unexpected, therefore, to see the attention that a number of researchers have paid to propose individual motivational constructs, namely “motivational self-guides” (Dörnyei 2014, p. 525), and the possible selves (Dörnyei 2005), in different models of learning styles, such as perceptual and social learning styles, and any other relevant variables experienced during the process of L2 learning.

To begin with, researchers have been interested in investigating the role of L2 motivation in the ultimate attainment of learners' language proficiency (e.g., Dörnyei 2005, 2014; Lanvers 2016). L2 motivation studies have observed a recent shift in focus in response to Gardner's (1985) concept of integrativeness and instrumentality with increasing attention paid to how learners' images of themselves as L2 learners might influence the learning process. Intending to develop a framework to accommodate diverse research trends within this recent shift, Dörnyei (2005) generated the L2 Motivational Self System, drawing on possible selves (Markus and Nurius 1986) and discrepancy theory (Higgins 1987). Dörnyei (2005, 2014) formulated and defined his framework as follows. The Ideal L2 self, central to the L2 Motivational Self System, refers to “the attributes that someone would ideally like to possess” (Dörnyei 2010, p. 78), the ought-to L2 self is defined as “the attributes that individuals believe they ought to possess to avoid possible negative outcomes” (Dörnyei and Chan 2013, p. 438), and the L2 learning experience “concerns situated ‘executive’ motives related to the immediate learning environment and experience” (Dörnyei 2009, p. 29).

The second variable worth investigating in L2 Willingness to Communicate (L2WTC) field of study is learning style which is comparatively consistent with individuals' information processing. It is defined as “a term that describes the variations among learners in using one or more senses to understand, organize, and retain experience” (Reid 1987, p. 89). To make a distinction between learning styles and other dimensions of behavior because of their similarities, Keefe (1979) regarded the former as “cognitive, affective, and physiological traits that are relatively stable indicators of how learners perceive, interact with, and respond to the learning environment” (p. 4). Kinsella (1995) opined that learning styles, regardless of teaching methods or content area, stay unchanged and fixed. The value of investigating learning styles lies in the fact that learning is achieved more effectively when information is presented in a way that matches learners' preferred method of acquiring and processing information

(Montgomery 1995). Researchers have attempted to categorize learning styles. Drawing upon the classifications of previous researchers, Reid (1987) defined perceptual learning styles as “a term that describes the variations among learners in using one or more senses to understand, organize and retain experience” (p. 89). She further classified perceptual learning styles into six categories of visual, auditory, kinesthetic, tactile, group, and individual (Reid 1998).

The third individual differences variable which has attracted attention from researchers and theorists is willingness to communicate, which has been found significant in differentiating language learners in their readiness and attempts to initiate talks with L2 speakers. Willingness to communicate (WTC) in EFL/ESL contexts is the construct explicating the differences in learners’ intention to communicate in the L2. Recently explored by a number of researchers (Ghonsooly et al. 2012; MacIntyre and Legatto 2011; Peng 2012), WTC is defined as “a readiness to enter into discourse, at a particular time with a specific person or persons, using L2” (MacIntyre et al. 1998, p. 547). Due to the importance of achieving a high level of language proficiency, WTC in a foreign language (L2WTC) has been found conducive to L2 use (MacIntyre et al. 1998), and it has been widely studied in relation to different individual, cognitive, and social psychological variables in ESL contexts (Cetinkaya 2005).

Motivated by the works provided above, and considering the imperative of achieving high level of language proficiency, identifying the contributing factors to learners’ L2WTC seems necessary. Therefore, this study aims to investigate the extent to which EFL learners’ WTC in English is predicted by L2 motivation self (Ideal L2 self in this study), perceptual learning styles (visual orientation in the present study), language proficiency, and self-perceived communication confidence among Iranian EFL learners. This study represents an innovative attempt to explore the prediction of the most recent L2 motivation theory coupled with psychological and linguistic variables on L2WTC. This research will also give us a deeper understanding of the WTC of Iranian English language major students, and help perceive the variables which can contribute to L2WTC in a single model. To this end, the constructs in addition to the formulated research questions are discussed consecutively in the following.

Review of Literature

Ideal L2 Self

The studies carried out by Gardner and his associates (e.g., Gardner 2000; Gardner et al. 2004) paved the way for identifying and recognizing the variables that are related to language learners’ motivation. Gardner’s socio-educational model proposed integrative and instrumental orientations as the two effective notions in language achievement. Integrative orientation, as the central component of the model, refers to the learners’ desire to learn a foreign/second language so as to identify and interact with the natives of that language and their culture. On the contrary, the instrumental orientation describes both “an open and positive regard for outside groups who speak L2,” which motivates individuals to learn a language due to factors such as better career prospects or higher credit ratings (MacIntyre et al. 2001, p. 373). A main tenet of Gardner’s

socio-psychological approach is that “the individual’s attitudes towards the second/foreign language and the L2 community, as well as his or her ethnocentric orientation in general, exert a directive influence on his or her L2 learning behaviour, which makes intuitive sense” (Dörnyei 2001, p. 48). Of the cognitive psychology theories, Deci and Ryan’s (2002) self-determination theory (SDT) classified motivation into three types, namely intrinsic motivation (IM), extrinsic motivation (EM), and amotivation based upon the inner interest of learners to participate in the activity (Dörnyei 2003). In a language learning context, IM refers to the attempts and efforts learners make in order to learn a language as a result of interest in the activity (Ellis 1997), while EM refers to the degree of the endeavors made to achieve an extrinsic reward (Vallerand 1997). Amotivation is defined as the lack of intent and intentionality to take action (Ryan and Deci 2000).

As a recent shift in focus, Dörnyei and his colleagues, drawing upon the framework of possible selves in social psychology (e.g., Markus and Nurius 1986) and “self-discrepancy” theory (Higgins 1987), proposed his L2 Motivational Self System (Dörnyei 2005, 2009), which regards integrativeness as a form of “psychological and emotional identification” (Dörnyei 2005, p. 96). This model consists of three components: the Ideal L2 self, the ought-to L2 self, and the L2 learning experience (Dörnyei 2009). The Ideal L2 self, defined as “the L2-specific aspect of one’s Ideal self” (Dörnyei 2005, p. 106), is deemed to motivate individuals to decrease the gap between their actual and ideal selves (Islam et al. 2013). On the contrary, the ought-to L2 self represents one’s vision of the attributes that one should have to meet his or her expectations and avoid any negative consequences. The L2 learning experiences, the third component, refer to the impact of the immediate learning environment and experience such as teachers and peers on the learning process (Dörnyei and Ushioda 2011). To make a connection between his model and already existing ones such as self-determination theory, Dörnyei (2005, 2009) related the constructs of Ideal L2 self to IM, and ought-to L2 self to EM.

The L2 Motivational Self System and its relevant issues have been investigated empirically in an extensive diversity of English as a foreign language (EFL) contexts, such as China, Japan, Indonesia, Iran and Pakistan (Islam et al. 2013; Lamb 2012; Papi 2010; Ryan 2009; Taguchi et al. 2009; Xu and Gao 2014), Hungary (Csizér and Kormos 2009), and Chile (Kormos et al. 2011).

Despite being at initial stages, L2 Motivational Self System has grasped substantial attention from SLA researchers. A majority of investigators (Asker 2012; Csizér and Kormos 2009; Islam et al. 2013; Kim 2009; Magid 2011; Magid and Chan 2012; Ryan 2009; Yashima 2009) have implemented research to validate the new framework in various parts of the world. Al-Shehri (2009) sought the interrelations among EFL learners in Saudi Arabia regarding their visual learning style, imagination, Ideal L2 self, and motivated L2 learning behavior. The study reported a positive correlation between learners’ visual learning preferences and their Ideal L2 self, which in turn would contribute to a high level of motivated L2 learning behavior. In another study conducted by Kim (2011) to investigate the relationships of Korean elementary school students’ perceptual learning style preferences, Ideal L2 self, ought-to L2 self, and motivated L2 learning behavior, the findings indicated that the students’ learning style preferences are closely related to the vividness of the Ideal L2 self-image and consequently to L2 learning behavior. Exploring the relation between L2 anxiety and L2

Motivational Self System, Papi (2010) found that owning a vivid Ideal L2 self and L2 learning experience can help learners lessen their L2 anxiety, while the ought-to L2 self remarkably worsens their anxiety.

Perceptual Learning Styles

Witkin's (1962) study, in classifying individuals into field independent (those taking analytic approach) and field dependent (those taking global predisposition) regarding information processing, is considered the start of research in the field of learning styles. From then on, many learning style dimensions have been proposed by various scholars. Asserting that there is an amalgam of at least 20 style dimensions, Oxford and Anderson (1995) singled out eight of them as particularly important for L2 learning. Among them, perceptual (sensory or physiological) preferences for processing information, comprising visual, auditory, kinesthetic, and tactile dimensions have attracted more attention. Visual style preference refers to activities such as reading and studying charts, auditory style to listening to lectures and tapes, kinesthetic style to experiential and total physical involvement in the activity such as role-playing, and tactile style to touching and manipulation of objects (Dörnyei 2005).

Drawing upon the already existing studies on perceptual modalities and perceptual learning style, Reid (1984) developed the Perceptual Learning Style Preference Questionnaire (PLSPQ). She added sociological aspects to develop a framework more feasible in ESL/EFL, hence forming six learning styles preferences: visual, auditory, kinesthetic, tactile, individual, and group.

Reid (1987) can be considered as the first to measure perceptual learning style preferences in L2 learning. She studied 98 English native and non-native speakers coming from different countries and majors. The findings revealed that ESL learners strongly preferred kinesthetic and tactile learning styles, while they preferred group learning the least. Furthermore, regarding majors, kinesthetic learning was a main learning style preference, but group learning was found to be perceived negatively except in computer sciences. The results indicated that males preferred visual learning styles, whereas females favored tactile learning. Despite the fact that no significant difference was seen in terms of age, it was found that “the older the student, the higher the preference means for visual, auditory, kinesthetic, and tactile learning” (p. 95).

Rossi-Le (1989) investigated the link between perceptual learning styles and learning strategies among 147 adult immigrants studying English in community college settings. The results reported that the participants strongly favored tactile and kinesthetic learning styles. Concerning gender, males, in comparison with females, substantially preferred tactile learning.

Wintergerst and DeCapua (2001) studied the learning style preferences of 32 undergraduate Russian-speaking learners studying ESL in the USA. The findings revealed that kinesthetic followed by auditory learning style was most favored by the participants. The interview analysis also showed that individual preferences were more important than cultural influences in learning style preferences.

Isemonger and Sheppard (2003) conducted a study to reveal the differences between language learners in Korea using the PLSPQ. Based on the results, a strong preference for kinesthetic learning, followed by auditory and tactile preferences, was observed. On

the contrary, the participants were found to least prefer individual learning. Female learners compared with the male ones indicated higher learning style preferences for kinesthetic and group learning styles. The other significant finding was that the students' style preferences with reference to age and major did not differ.

In a study to investigate the links between the perceptual learning style, Ideal L2 self, and motivated L2 behavior of Chinese, Japanese, South Korean, and Swedish high school students, Yang and Kim (2011) reported that perceptual learning styles (visual, auditory, kinesthetic styles) were meaningfully correlated with their Ideal L2 self and motivated L2 behavior, but only the Ideal L2 self was identified as a meaningful predicting factor for their motivated L2 behavior. In Chinese learners, visual and auditory learning styles were correlated with Ideal L2 self and motivated L2 behavior. In comparison with other countries, Chinese learners were found to indicate a higher level of motivated L2 behavior, but Swedish learners showed a higher level of Ideal L2 self than the Chinese ones. Yang and Kim (2011) put contextual influences as the reasons, such as sociocultural environment among the learners of the study.

Concerning the importance of imagery-related capacity, You et al. (2016) conducted a study to incorporate visualization in second language (L2) motivation theory. The results showed that mental imagery significantly contributes to the motivational disposition of Chinese EFL learners. It was argued that vivid imagery is strongly linked to Ideal L2 self, meaning that future self-guides are fueled by one's vivid vision.

WTC

Originally explored in an L1 context of communication, WTC is considered as a personality trait and justifies the reason for which individuals show similar tendencies in different communication contexts (McCroskey and Baer 1985). However, when WTC comes to EFL/ESL, it is a different concept, as MacIntyre et al. (1998) mentioned that it is virtually impossible to equate first language willingness to communicate (L1WTC) with second language willingness to communicate (L2WTC). Therefore, they developed a model for L2WTC integrating psychological, linguistic, and contextual variables to investigate the dual characteristics of trait and state in L2WTC, which is different from the trait feature of WTC in L1.

Many studies indicated interweaving relationship between L2WTC and other individual variables namely with motivation and attitudes (Ryan 2009), learner's beliefs (Peng and Woodrow 2010), self-confidence (Clément et al. 2003), personality (Ghonsooly et al. 2012), and age and gender (MacIntyre et al. 2002).

Peng and Woodrow (2010) examined a hypothesized model comprising WTC in English, communication confidence, learner's beliefs, motivation, and classroom environment using structural equation modeling (SEM). The findings revealed that motivation affects communication confidence but did not influence directly WTC. Peng (2014) conducted a study to explore the interrelationships between the motivational self-system, international posture, L2 anxiety, and WTC using structural equation modeling. The results indicated that L2WTC inside the classroom was influenced by L2 anxiety and L2 learning experience directly. He also found that Ideal L2 self and ought-to L2 self, respectively, had a negative and positive effect on L2 anxiety and in turn L2 anxiety has a negative effect on in-class L2WTC.

L2 Confidence

L2 confidence, also called communication confidence or L2 self-confidence, referring to the interrelationship between the individual and the L2, is discussed as the ability of language learners to effectively and adaptively communicate in an L2. McCroskey (1984) defines communication competence as “adequate ability to pass along or give information; the ability to make known by talking or writing” (p. 261). L2 confidence is different from other kinds of confidence in that it corresponds to situations in which L2 is concerned, not sports or any other academic subjects (MacIntyre et al. 1998). L2 confidence comprises two components of self-evaluation of L2 skills, the cognitive aspect, also known as Self-Perceived Communication Competence (SPCC), and language anxiety (Horwitz et al. 1986), and the affective aspect which is also known as communication apprehension. Findings have revealed that these two constructs are highly correlated to each other, thereby deserving to be defined as a latent trait of L2 self-confidence (Clément 1986). According to Lockley (2013), SPCC has many potential implications for second/foreign language learning and teaching as one of the major factors in predicting WTC. Talking about SPCC, it can be considered as how an individual perceives her/his own competence for spoken communication in a certain context. This is an important area of research as SPCC can have a strong impact on WTC that in turn can affect FLL to a great deal.

L2 self-confidence was proved to be correlated to aspects of intergroup contact (Clément and Kruidenier 1985) and actual competence in the L2 (Clément 1986), as well as ethnic identity (Clément and Noels 1991), intercultural adaptation (Noels et al. 1996), motivation and L2WTC (Cetinkaya 2005; Ghonsooly et al. 2012; Peng and Woodrow 2010; Yashima 2002), Ideal L2 self (Khajavy and Ghonsooly 2017), and L2 proficiency (Ghonsooly et al. 2012).

Hypothesized Model and Significance of the Study

Based upon former studies and earlier data analyses, a model of L2WTC was hypothesized to integrate five variables: WTC in English, English language achievement, perceptual learning styles (visual style in this study), communication confidence, and L2 motivation (Ideal L2 self in the present study). The initial model was hypothesized based on the following considerations.

Following the empirical study of Kim and Kim (2014) and that of Al-Shehri (2009), we postulated a direct path from perceptual learning style to Ideal L2 self. Also, as the studies conducted by Kim and Kim (2011) and Dörnyei and Chan (2013) confirmed, a direct path was added from Ideal L2 self to language proficiency.

MacIntyre et al.'s (1998) pyramid model indicates an indirect effect of motivation on L2WTC. Recent empirical research (Cetinkaya 2005; Ghonsooly et al. 2012; Peng and Woodrow 2010; Yashima 2002) revealed an influence of motivation on L2WTC through communication confidence. Accordingly, we have drawn a path from Ideal L2 self (as a subset of the new theory of Motivation L2 System developed by Dörnyei 2005) to communication confidence.

As shown by MacIntyre et al.'s (1998) pyramid model, L2 proficiency is among the factors that indirectly influences L2WTC. Following Gardner et al. (1997) and Khajavy's et al. (Khajavy et al. 2014), L2 achievement is a very strong predictor of communication confidence. Thus, we have drawn a path from L2 achievement to communication confidence.

This present study, however, did not assume a direct effect of Ideal L2 self on L2WTC, because motivational variables have been widely found to indirectly affect L2WTC through their effects on other more proximal factors such classroom environment or confidence (MacIntyre et al. 1998).

Also, studies on L2 learning styles have revealed that perceptual learning styles can be indirectly associated with L2 achievement. Such a possible relationship between learning styles and language proficiency can be attained by the intervening effect of motivated behavior. Figure 1 illustrates the structural relationship hypothesized in this study.

Objectives of the Study

Since Dörnyei's (2005, 2009) L2 Motivational Self System is a relatively new framework in the field of L2 motivation, the relationships between motivation and other effective variables on L2WTC have not been sufficiently investigated. Given this, this study is an attempt to test the proposed model integrating Ideal L2 self, communication confidence, perceptual learning styles, and language proficiency on L2WTC. To this aim, the following research question was formulated:

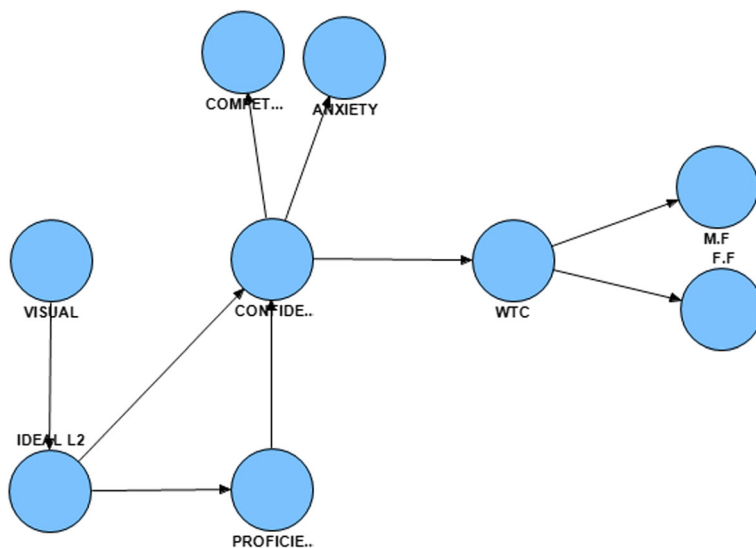


Fig. 1 The postulated relationships between Ideal L2 self, perceptual styles, L2 confidence, proficiency, and L2WTC

Research Question 1

Is the proposed model appropriate for Iranian English major students?

The literature offers support for the following hypotheses:

H1; perceptual learning style indirectly and positively affects language proficiency through the mediation of Ideal L2 self.

H2; Ideal L2 self positively and directly affects L2 confidence, which in turn affects L2WTC.

H3; L2 self-confidence mediates the effects of language proficiency on L2WTC.

Methodology

Participants and Data Collection Procedures

A total of 150 university freshmen participated in this study, including 90 females (60%), 60 males (40%). This number of participants was selected according to *Ten Times Rule* (i.e., 10 times the number of independent variables of the most complex OLS regression in the structural or formative measurement model) (Hair et al. 2014). Participants were randomly selected from three universities in Tehran, Iran. All participants had passed a very challenging university entrance exam, called Kunkor, and all of them were studying English as an academic major and had upper-intermediate language proficiency level. The age range of the participants was 18–25; the mean age was 22.39 ($SD = 2.6$). The universities in which the study was supposed to be conducted as well as the teachers were informed of the purpose of the study. To ensure that the learners do not answer the questions haphazardly, they were told that it is optional to participate in the study. It took the researcher approximately 1 month to complete all of the questionnaires.

Instruments

WTC in English

Adapted from Weaver (2005), ten items, on a 7-point Likert scale, were employed to measure WTC in meaning-focused and form-focused English activities.

L2 Confidence

Found as the most significant predictor of L2WTC (Cetinkaya 2005; Ghonsooly et al. 2012), L2 confidence was defined by MacIntyre et al. (1998) as the “overall belief in being able to communicate in the L2 in an adaptive and efficient manner” (p.551). This construct has been divided into two dimensions of perceived competence and foreign language anxiety (Clément 1980). To this end, six items from Peng and Woodrow (2010, adapted from Weaver 2005) for Perceived Communicative Competence in English, and ten items from Horwitz et al. (1986), translated and validated by Khodadady and Khajavy (2013) in the Iranian context, were employed to measure communication anxiety in English.

Ideal L2 Self

The current study used the framework established and developed by Dörnyei's (2005, p.105) L2 Motivational Self System, which is made up of three dimensions: the Ideal L2 self, ought-to L2 self, and L2 learning experience. This study is an attempt to explore the relationships between Ideal L2 self and other variables, as mentioned in the background. The internal consistency of this instrument was also measured using Cronbach's alpha, and it was $\alpha = .89$ in this study. The questionnaire, consisting of six items on a 6-point Likert scale in the Iranian version, has been employed for the purpose of the current study (Dörnyei 2009). This includes items such as "*I can imagine myself speaking with international friends and colleagues.*"

Perceptual Learning Styles

The PLSPQ developed by Reid (1984) is regarded as the first instrument to measure non-native EFL/ESL perceptual learning styles. It is comprised of thirty randomly ordered statements for six learning styles (five statements on each learning style) on a 5-point Likert scale: visual, auditory, kinesthetic, tactile, individual, and group. As Reid (1987) stated, "validation of the questionnaire was done by the split-half method. Correlation analysis of an original set of 60 statements (ten per learning style) determined which five statements should remain within each subset" (p. 92). Later on, she classified styles as major, minor, or negligible (or negative). However, for the research purpose of the current study, the researcher considered just the visual learning style in the study as previous studies have found correlations between this style and Ideal L2 self.

Language Proficiency

Iranian students must pass university entrance exams to pursue their academic studies at universities and academies. The entrance exam, also called Kunkor, assesses the test takers general knowledge in fields such as *Literature*, *Mathematics*, and in particular *General English*. Since all the learners take the same test under rather similar circumstances, the study intends to take their score in the *General English* section as the construct of language proficiency.

Statistical Analysis for Measurement

The collected data were first inputted into the Statistical Package for the Social Sciences (SPSS) 20 software prior to being subjected to SEM analysis using SmartPLS. All model parameters were estimated via maximum likelihood estimation.

PLS-SEM, consisting of measurement and structural models, works efficiently with small sample sizes and complex models and makes practically no assumptions about the underlying data in terms of data distributions (Hair et al. 2014). Furthermore, it can simply handle both reflective and formative measurement

models. In addition, single-item constructs can be easily run with no identification problems (Hair et al. 2014). The other worth mentioning benefit of PLS-SEM is its greater statistical power compared with covariance-based SEM, reporting a specific relationship significant when it is in fact significant in the population. Considering all the pluses of the software, the researchers have decided to apply SmartPLS for the intended aims.

Results

To analyze data by means of partial least squares, Hulland (1999) proposes a model of two stages: measurement model and structural model. Each consists of a definite set of criteria which will be further elaborated on in the following sections.

Measurement Model

Assessment of reflective measurement models includes composite reliability to evaluate internal consistency, individual indicator reliability, and average variance extracted (AVE) to evaluate convergent validity. Also, discriminant validity is obtained using the Fornell-Larcker criterion and cross loadings (Hair et al. 2014). Table 1 provides statistics about each criterion of measurement model (Fig. 2).

The reliability estimates of each variable computed via Cronbach's alpha are as follows: anxiety ($\alpha = .8$), competence ($\alpha = .84$), confidence ($\alpha = .9$), Ideal L2 self ($\alpha = .8$), form-focused WTC ($\alpha = .9$), meaning-focused WTC ($\alpha = .9$), visual learning style ($\alpha = .91$), and WTC ($\alpha = .92$), each showing an acceptable level of internal consistency for all variables.

Additionally, SmartPLS provides researchers with the ability to apply a different measure of internal consistency, rather than just individual reliability, to compensate for the limitations of Cronbach's alpha. The values between 0.7 and 0.9 can be regarded as satisfactory in this type of reliability (Nunnally and Bernstein 1994). Table 1 clearly represents the values that are quite acceptable within the range.

Table 1 Estimates of reliability and validity

Total	AVE	Composite reliability	R square	Cronbach's alpha
Anxiety	0.552893	0.860377	0.899867	0.796998
Competence	0.668109	0.885220	0.936498	0.841850
Confidence	0.514395	0.919650	0.525658	0.902772
Language proficiency	1.000000	1.000000	0.407617	1.000000
Form-focused	0.893283	0.890981	0.800341	0.900051
Ideal L2	0.538812	0.869950	0.629566	0.819140
Meaning-focused	0.837714	0.898674	0.911377	0.910935
Visual	0.742780	0.905174		0.913725
L2WTC	0.742276	0.910131	0.197275	0.920925

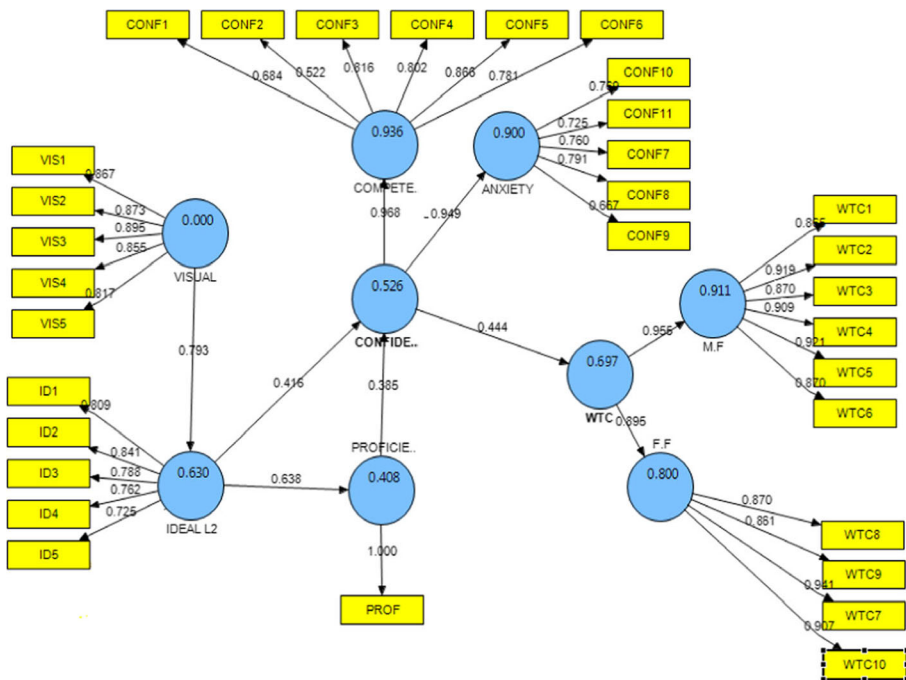


Fig. 2 Factor loadings of latent unobserved variables

Regarding convergent validity, the extent to which a measure correlates positively with alternative measures of the same construct is measured by AVE (average variance extracted) (Hair et al. 2014). As shown in Table 1, all the values are higher than .5, signifying an appropriate estimate for the variables.

Furthermore, the proposed model complies with discriminant validity (Table 2) as all the square root of the AVE of each construct is higher than its highest correlation with any other construct (Fornell and Larcker 1981).

Table 2 Cross-loading correlations

Total	Anxiety	Communicative competence	Language proficiency	Form-focused	Ideal L2 self	Meaning-focused	Visual
Anxiety	0.7416198						
Communicative competence	0.738784	0.748331					
Language proficiency	0.608289	0.627735	1.000000				
Form-focused WTC	0.245700	0.293410	0.406427	0.943398			
Ideal L2	0.611458	0.648753	0.638449	0.326769	0.728010		
Meaning-focused WTC	0.486619	0.460555	0.764708	0.721039	0.537661	0.911043	
Visual	0.620874	0.655824	0.650739	0.376841	0.793452	0.534126	0.861510

Structural Model

Having confirmed the reliability and validity of the constructs, it is time to investigate the structural model of the proposed scheme. This model enables researchers to represent the underlying theory/concept of the path model to determine how well the theory/concept can be confirmed via empirical data. To this end, the key measures, such as significance of the path coefficients, the level of the R^2 values, the f^2 effect size, and the predictive relevance (Q^2) size, are analyzed.

At this stage, the researcher has run bootstrapping to compute the empirical t values associated with each parameter. As Fig. 3 indicates, all the path coefficients between the latent variables are significant in which the critical value for two-tailed tests is 1.96.

Regarding the coefficient of determination (R^2), a measure of the model's predictive accuracy, the squared correlation between a specific endogenous construct's actual and predicted values is calculated. R^2 , as the most commonly used measure of structural model, represents the exogenous latent variables' combined effects on the endogenous latent variable. Its value ranges from 0 to 1; however, the values of 0.75, 0.50, or 0.25 for endogenous latent variables can be respectively described as substantial, moderate, or weak (Hair et al. 2014). Its predictive accuracy increases as the value goes toward one. The results show that these are significant and all exceeded 0.4. As an example, 70% of variances in L2WTC can be predicted by self-confidence.

In addition to assessing the R^2 values of all endogenous constructs, effect size (f^2) can be employed to evaluate the impact of a specific exogenous construct on endogenous constructs when omitted from the model. This measure can be used just for L2 self-confidence in the proposed model in Fig. 1 because other endogenous variables (Ideal L2 self, proficiency, and L2WTC) are influenced by one exogenous variable. The effect size (f^2), according to Cohen (1988), is .43, which implies a large effect of exogenous variables.

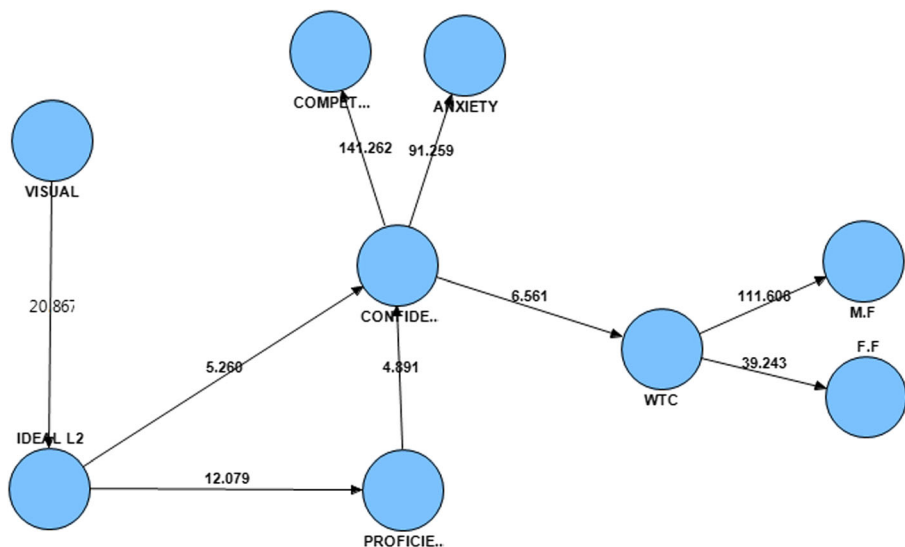


Fig. 3 Path coefficients of latent variables

Table 3 Stone-Geisser's Q^2 value

Total	1-SSE/SSO
Anxiety	0.478841
Communicative competence	0.519176
Confidence	0.258209
Language proficiency	0.404250
Form-focused WTC	0.725014
Ideal L2	0.326255
Meaning-focused WTC	0.751304
L2WTC	0.539858

Researchers also examine Stone-Geisser's Q^2 value (Geisser 1974; Stone 1974), besides the coefficient of determination, to measure the model's predictive relevance. This is done by resampling procedure, also known as bootstrapping, and is applied just for reflective models, not the formative ones. In the structural model, Q^2 values greater than zero prove the path model's predictive relevance for a certain reflective endogenous latent variable. As provided in Table 3, all the latent variables exhibit a significant level of predictive relevance (higher than 0.3).

The last statistical procedure which has been administered in the study relates to goodness of fit (GOF). The calculated index of GOF for the current study is 0.67, representing a good criterion to confirm the model (Wetzels et al. 2009). Thus, the estimated model has demonstrated adequate fit to the data, as indicated by the GOF index.

Discussion

The data analysis revealed that the hypotheses and research questions raised in the present study were confirmed. The results are in line and in contrast with several studies that will be elaborated on in the following sections.

Appropriateness of the Model Proposed for Iranian English Major Learners

To answer the first raised research question, the statistical procedure known as GOF has been administered. The calculated index of GOF for the current study is 0.67, representing a good criterion to confirm the model (Wetzels et al. 2009). Thus, the estimated model has demonstrated adequate fit to the data, as indicated by the GOF index, confirming that the proposed model is appropriate for the Iranian EFL learners.

This finding can be explained in light of the fact that visually oriented language learners in Iran are more inclined to spot the difference between their present and ideal level, and have a clearer imagination of what to be in the future. Such capability makes language learning purposeful and attainable; this in turn leads to a higher level of self-confidence. Awareness of the distance between their actual and Ideal L2 selves, and encouragement from a high level of confidence motivate language learners to put further effort and get more willing to engage in communication with L2 speakers. This is in line with the studies of Huang et al. (2015) and Ryan and Dörnyei (2013) in which possible selves were found to be strong predictors of language learners entering communication

with the L2 speakers. Also, this result corroborates that of Khajavy et al.'s (2014) which reported the role of language attainment and self-perceived communication confidence in determining language learners' levels of WTC with others.

Relation between Visual Learning Style, Ideal L2 Self, and Language Proficiency

Reid (1987) defined perceptual learning styles as the characteristics of language learners to which their variations can be attributed. These cognitive, affective, and psychological attributes can stay fixed, remain unchanged (Keefe 1979), and are regarded as the preferred style of learners for acquiring and processing information (Montgomery 1995). Visual learning style, or learners' orientations to rely on their eyesight and vision, has been investigated in the current study.

The results revealed that the visual learning style has a positive and direct effect on Ideal L2 self, and is indirectly associated with L2 proficiency through the mediating function of Ideal L2 self. The reasons can be related to the fact that visually oriented learners draw on their visualization potentiality and their capacity to employ mental imagery to develop a more vivid image of their future selves. Aware of the distance between their actual and Ideal L2 selves, learners become encouraged to make more endeavors to reach higher levels of proficiency. It can be argued that learners' thinking styles (i.e., visual style) make learning a goal-oriented and purposeful activity, as learners become cognizant of their ideals by their imagination, which motivate them to enhance their L2 performance and fill the gap between the two selves. When motivated, they put further effort into getting more input and exercise language activities more to approximate their desired proficiency level.

The result is consistent with the empirical studies of Al-Shehri (2009) and Kim (2009) which have reported that visually oriented language learners are more inclined to create a more vivid imagination of their selves, thereby more likely to develop a more potent Ideal L2 self. Al-Shehri (2009) mentioned that this learning style enables learners to "have a better chance of imagining more vivid Ideal selves, which in turn will generate stronger motivation to attain such Ideal selves" (Al-Shehri 2009, p. 166). This is also in line with Dörnyei's (2009) research which has indicated that Ideal L2 self, leading learners to form a self-image, results in learners' higher engagement in L2 learning behavior. Moreover, Ryan and Dörnyei (2013) have made the claim that Ideal L2 self is the most effective motivator of the learning process and language success (Huang et al. 2015; Lanvers 2016).

Relation between Ideal L2 Self, L2 Self-Confidence, and WTC

Dörnyei (2005) defined L2 confidence as "personal's beliefs in ones' own ability to produce results, accomplish goals or perform task competently" (p. 73). Clément's (1980, 1986) proposed model of L2 self-confidence contains perceived competence and lack of anxiety. The data analysis identified that Ideal L2 self can positively and directly affect the variations of language learners' self-confidence.

It can be hypothesized that L2 learners able to draw on their imaginative capacity and perceive a vivid image of their ideal selves seem to conform their aspirations to what is expected in L2 contexts. When they imagine themselves as competent users of English, they turn more confident. For example, when learners imagine themselves

Table 4 Research hypotheses

Hypotheses	Path coefficients	<i>t</i> values	Significance levels
Visual→Ideal L2 self	.79	20.86	Significant
Ideal L2 self→language proficiency	.63	12.09	Significant
Ideal L2 self→L2 self-confidence→L2WTC	.6	6.01	Significant
Language proficiency→L2 self-confidence→L2WTC	.2	3.09	Significant

talking to friends in the L2 or giving a lecture in academic seminars, they become confident of their abilities. The desire to bridge the gap between their current and ideal selves would help learners reduce anxious arousals and feel more competent to accomplish L2 tasks, and as a result L2 self-confidence enhances. This intricate relationship between Ideal L2 self and self-confidence can be ascribed to the fact that both capitalizes on individuals' conceptions of their abilities.

The other significant finding of the present research relates the positive and indirect predicting role of Ideal L2 self in WTC through the mediation of L2 self-confidence. Learners' conceptions of competence to convey their meanings and get their aims done enable them to have positive evaluations of own skills and feel less worried (i.e., higher self-confidence), motivating learners to enter into discourse to communicate with particular speakers.

This is in line with the studies of Khajavy and Ghonsooly (2017) and Munezane (2016) which reported that L2 self-confidence can positively and directly predict L2WTC. Besides, Peng (2014) found that self-confident learners tend to communicate with others. Consequently, to put it another way, language learners' imaginary representation of their confidence, which can bring about less anxiety, would make them more prepared and ready to participate in conversations and discourse.

Relation between Language Proficiency, L2 Self-Confidence, and L2WTC

The other objective of this study was to explore the relation between language learners' proficiency and their WTC with others. As the results have clearly represented, language proficiency positively and indirectly affects L2WTC through the intervening role of self-confidence.

In principle, it can be justified in this light that when language learners get motivated by perceiving a vivid imagination of their ideal selves, they exert more effort to more effectively and efficiently perform their L2-related activities. Observing their gradual progress at each learning stage, learners feel a high sense of competence and low level of anxiety, and thus more self-confident. They come to know that making more effort will provide them with more opportunities to learn and more awareness of the rich domain of their abilities. Learners' beliefs in their abilities to accomplish L2 tasks motivate them to be more willing to communicate and enter into discourse with others. Such a relationship was reported by MacIntyre's (MacIntyre et al. 1998) study in which an indirect effect of L2 proficiency on L2WTC was identified.

To interpret the results, Table 4 was used to analyze the research questions. As clearly presented, all the questions were accepted.

Conclusion and Implications

The current study provides some substantial findings on the relation between language learners' Ideal L2 self, as a critical component of Dörnyei's (2005) L2 Motivational Self System, visual learning styles, L2 self-confidence, language proficiency, and L2WTC. Overall, the results have shown that visually oriented language learners tend to employ more imaginative capacity and mental images, thereby having greater estimation of their ideal selves. This finding is in line with the empirical studies of Al-Shehri (2009) and Kim (2009) which have reported similar relations between the two mentioned variables. Also, the research contributes to the literature that Ideal L2 self is a significant predictor of L2WTC by the mediating effect of L2 confidence. It has been explained in this light that motivated language learners feel less aroused by worrying factors, and more confident of their abilities to competently finish intended tasks, offering validity evidence for previous studies. Additionally, the study adds to the current literature on L2WTC by showing that proficient language users, as a result of feeling more communicatively competent and less anxious, tend to enter into discourse with other speakers and communicate with them. This result is consistent with MacIntyre's (MacIntyre et al. 1998) study in which an indirect effect of L2 proficiency on L2WTC was reported.

The findings have significant implications for language teachers to motivate their learners to ultimately enhance their performance in L2-use situations and encourage them to further participate in interactions with other speakers. Teachers and instructors are recommended to know that visually oriented language learners draw on vision and imagination to bridge the gap between their actual selves and Ideal L2 ones. Knowing the distance between their selves, learners exert a greater amount of effort to more effectively and efficiently perform in the L2-related settings, as confirmed by Dörnyei (2009). Also, language teachers and class managers are expected to enable their learners to identify their future selves which can help them be more confident of their capabilities to accomplish tasks, and their capacity to fulfill their objectives. This in turn makes them more willing to communicate with particular individuals in the L2. Therefore, teachers should help learners to recognize the gap between their actual and ideal level of language proficiency, which can inspire them to have more interactions in L2 activities.

The current study, like any other ones, has some limitations. First, the information about language learners' test scores was self-reported. Obtaining information in such a way can involve dishonest and untrue data which might invalidate the findings. This can be regarded as one of the limitations of the study. Second, there has been no attempt on the side of researchers to control the effect of gender-related variance as Henry and Cliffordson (2013) remarked that males and females are likely to develop possible selves. Thus, gender differences can be a potential threat to generalize the findings of the study. The third shortcoming would be related to the selection of Ideal L2 self, as one of the critical components of L2 Motivational Self System (Dörnyei 2005). Future research can include other aspects of the theory, i.e., ought-to L2 self and L2 learning experience, to find their relations with other individual variables. Besides, other learning styles, namely auditory and kinesthetic styles, can be considered in the following studies to investigate the effects on the

other subcomponents of L2 Motivational Self System. Fourth, almost all the language learners who participated in the study had upper-intermediate level of language proficiency, which can make it difficult to generalize the findings to other contexts in which the participants are of higher or lower language proficiency. The researchers interested in this field of work are suggested to replicate the current study on learners of different language proficiency levels.

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